08/2026 Ex:04 LAN CONNECTIVITY.

AIM:

To setup and configure a LAN using a switch and Ethernet cables in your lab.

What is a LAN?

↳ A Local Area Network refers to a Network that connects devices within a limited area, such as an office building, school, or home. It enables users to share resources, including data, printers and internet access.

PROCEDURE:

1. Plan and Design an appropriate network topology taking into account network requirements and equipment location.

2. You can take 4 Computers, a Switch with 8, 16, or 24 ports which is sufficient for networks of these sizes, and 4 Ethernet cables.

3. Connect your computers to networks switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet cable into your computer and the other end into your network switch.

4. Assign IP address to your PC's

(a.) Log on to the client comp as Admin

(b) Click Network and Internet Connections.

(c) Right Click Local Area Network connection → Go to Prop's → select Ip → Click on Prop's → select use the following ip address option and assign ipaddress.

5. Configure a network switch.

6. Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

7. Select a folder → go to prop's → click sharing tab → share it with everyone on the same LAN.

8. Try to access the shared folder from others Computers of the network.

RESULT:

Successfully configured a LAN using a switch, assigned IP addresses to all devices, verified connectivity using the ping command, and shared files b/w the connected computers.